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LLaSM: Large Language and Speech Model

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Abstract

Multi-modal large language models have garnered significant interest recently. Though, most of the works focus on vision-language multi-modal models providing strong capabilities in following vision-and-language instructions. However, we claim that speech is also an important modality through which humans interact with the world. Hence, it is crucial for a general-purpose assistant to be able to follow multi-modal speech-and-language instructions. In this work, we propose **L**arge **L**anguage **a**nd **S**peech **M**odel (**LLaSM**). LLaSM is an end-to-end trained large multi-modal speech-language model with cross-modal conversational abilities, capable of following speech-and-language instructions. Our early experiments show that LLaSM demonstrates a more convenient and natural way for humans to interact with artificial intelligence. Specifically, we also release a large Speech Instruction Following dataset LLaSM-Audio-Instructions. Code and demo are available at `https://github.com/LinkSoul-AI/LLaSM` and `https://huggingface.co/spaces/LinkSoul/LLaSM`. The LLaSM-Audio-Instructions dataset is available at `https://huggingface.co/datasets/LinkSoul/LLaSM-Audio-Instructions`.

1 Introduction

Speech contains semantic information and contains paralinguistic information like intonation at the same time, it carries more quantity of information than text. Additionally, speech is a more convenient and natural way for humans to interact with artificial intelligence. Therefore, following speech-and-language instructions is crucial when developing a general-purpose assistant.

However, most large language models [1, 2, 3] receive text input only, which restricts the ability of large language models. Vision-and-language multi-modal models [4, 5, 6, 7, 8, 9] offer the ability to understand the vision information, making a huge step toward general artificial intelligence (AGI), but it is still inconvenient for humans to input the tasks by typing a text instruction. The cascading paradigm methods [10, 11] use an automatic speech recognition (ASR) model to convert the speech input into the text input, then the model can process the task with the text input. However, it still leads to information consumption during the modal transformation from speech to text and might import mistakes of the ASR system. Recently, speech-language multi-modal models [12, 13] focusing on processing and generating speech and text with a large language model are capable of understanding and generating multi-modal content. The speech signals are encoded into discrete tokens, and then discrete speech tokens are expanded into the vocabulary of the LLM. In this way, the LLM needs to be retrained with plenty of multi-modal data and huge computing resources.

In this paper, we propose LLaSM, a large speech-and-language model with cross-modal conversational abilities, capable of understanding and following speech-and-language instructions. Following the manner of LLaVA [6], we leverage the well-trained speech modal encoder and the LLM, which makes LLaSM more resource-friendly. Specifically, we use Whisper [14] as a speech encoder to encode the speech signals into embeddings. Then a modal adaptor learns to

align speech embeddings with the input text embeddings of the large language model. The speech embeddings and the text embeddings are concatenated together to form interleaved sequences, then the interleaved sequences are input to the LLM for supervised fine-tuning. The training process is divided into two stages. In the first stage, we use the public ASR datasets for the modality adaptation pre-training. The speech encoder and the LLM are frozen, only the modal adaptor is trained to align the speech and text embeddings. As most of the model parameters remain frozen, only a small part of the parameters from the modal adaptor is trained during this stage, it is not resource-consuming. In the second stage, we use cross-modal instruction data for training to provide the model with the capacity to process cross-modal conversations and handle multi-modal instructions. The speech encoder is frozen while the parameters of the modal adaptor and the language model are updated for cross-modal instruction fine-tuning. Worth noting that existing open-source speech-text cross-modal instruction-following datasets are scarce, so we build and release a speech-text cross-modal instruction-following dataset **LLaSM-Audio-Instructions**. The dataset is constructed by carefully selecting dialogues from GPT4-LLM [15], ShareGPT [16], WizardLM [17], and using text-to-speech technology to generate a large amount of dialogue audio data. In total, it contains 199k conversations, in which there are 80k Chinese audio samples and 428k English audio samples, which is the largest Chinese and English speech-text cross-modal instruction-following dataset to our knowledge.

Our paper makes the following contributions:

- We build a speech-language multi-modal model that can understand and follow speech-language instructions, which provides a more convenient and natural way for humans to interact with artificial intelligence.
- We construct and release LLaSM-Audio-Instrustions, a large-scale Chinese and English speech-text cross-modal instruction-following dataset. We release the data in `https://huggingface.co/datasets/LinkSoul/LLaSM-Audio-Instructions`.
- We release the code in `https://github.com/LinkSoul-AI/LLaSM` and the demo is shown in `https://huggingface.co/spaces/LinkSoul/LLaSM`.

2 Related Work

Vision Large Language Model has gained significant traction [4, 5, 6, 7, 8, 9] recently. Most of them leverage the pre-trained LLMs and vision encoders to perform vision tasks. Flamingo [18] aligns a pre-trained vision encoder and language model using gated cross-attention and is trained on billions of image-text pairs. BLIP-2 [19] employs a Flan-T5 [20] with a Q-Former to efficiently align visual features with the language model. Palm-E [5], featuring 562 billion parameters, integrates the 540B PaLM [2] and 22B Vision Transformer [21] into the largest vision-language model. LLaVA [6] leverages pre-trained CLIP [22] visual encoder and LLaMA [3] and conducts instruct tuning on GPT4-assisted visual instruction data. GPT-4 [4] also shows powerful visual understanding and reasoning abilities. The success of the multi-modal large language model in the visual domains has brought a lot of inspiration to the research in the speech domains as well.

Speech Large Language Model has gained more and more interest, for the success of the vision multi-modal LLMs. The cascading paradigm methods [10, 11] use an automatic speech recognition (ASR) model to convert the speech input into the text input, which still leads to information consumption and might import mistakes of the ASR system. Recently, speech-language multi-modal models [12, 13] focusing on processing and generating speech and text with a large language model are capable of understanding and generating multi-modal content. The speech signals are encoded into discrete tokens, and then discrete speech tokens are expanded into the vocabulary of the LLM. In this way, the LLM needs to be retrained with plenty of multi-modal data and huge computing resources.

3 Approach

3.1 Model

The focus of training multi-modal models is to fuse cross-modal complementary information of multi-modalities and effectively exploit the capabilities of well-trained large language models. The LLaSM model architecture is shown in Figure 1. We use Whisper [14] to encode the raw audio data into embeddings first, then a modal adaptor is trained during the pre-training stage to align the audio embeddings and the text embeddings. The audio embeddings and the text embeddings are concatenated together to form interleaved input sequences to input to the large language model. We choose Chinese-LLAMA2-7B [23] as our LLM, for its capabilities in both Chinese and English. During the cross-modal instruction fine-tuning stage, the modal adaptor and the LLM are trained with multi-tasks.

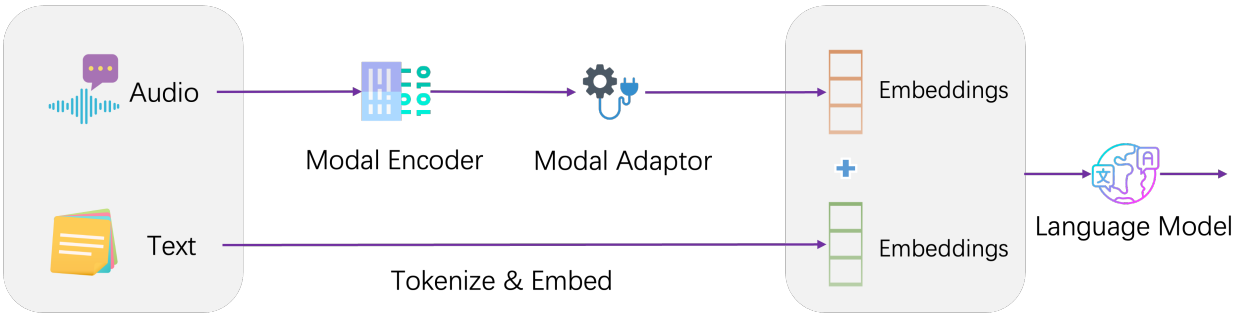

Figure 1: Model framework of the LLaSM

The pre-training stage. During this stage, the modal encoder and the LLM remain frozen. To enable the LLM to understand the audio embeddings from the modal encoder, the modal adaptor is trained with public ASR data to align the text and the audio embeddings. The data sample (audio data, text label) of ASR data is formatted as a tuple of (simple instruction, audio data, text label), in which the simple instruction is an automatic speech recognition instruction. According to the different languages of the audio data, an English simple instruction listed in Figure 2 or a Chinese simple instruction listed in Figure 3 will be chosen. The unified format of the pre-training multi-modal sequence X_{sample} is shown in Figure 4. Each data sample is formatted as X_{sample}, then we will replace the audio patch embeddings from the text sequence with the audio embeddings of the modal adaptor. The final interleaved input embeddings will be input to the large language model. The training target is to predict the text label of each data sample.

- "Transcribe the following speech into text."
- "Convert the spoken words into written text."
- "Transform the speech into a written transcript."
- "Record the oral communication as written text."
- "Render the verbal message into written form."
- "Transpose the spoken language into written script."
- "Translate the spoken words into written text."
- "Write down the words spoken in the speech."
- "Transform the spoken discourse into a written document."
- "Convert the spoken utterances into written transcription."
- "Create a written record of the spoken speech."

Figure 2: English simple instructions.

- "请将以下语音转译为文本"
- "请将这段语音转换成文字。"
- "能否把这段语音转成文字形式？"
- "把这段语音转写成文字。"
- "将此录音转化为文字。"
- "把这段音频转换成文字。"
- "能否把这段录音转成文字？"
- "把这段语音转化为文本。"
- "将这段录音转为文字。"
- "把这段语音转为文字格式。"
- "能否将这段录音转换成文本？"

Figure 3: Chinese simple instructions.

The cross-modal instruction fine-tuning. During this stage, only the modal encoder is frozen, the modal adaptor and the LLM are joint-trained with multi-tasks. We build complex cross-modal instructions using several conversational data. The questions from humans are generated to audio data by using Microsoft Azure text-to-speech API, then the training target is to predict the responses from the chatbot. A round of question and answer will be processed into a multi-modal sequence X_{sample}, and multiple rounds of question and answer will be concatenated with the EOS token. The unified format of the cross-modal instruction fine-tuning sequence is shown in Figure 5. As the effectiveness of text-only conversational data with multi-task instructions has been demonstrated in several open-source language-only instruction-tuning works [15, 16, 17], the cross-modal instructions are able to improve the capacity of following multi-modal instructions.

3.2 Data Collection

To enable the LLM to understand the audio signals, we collect several public ASR data sets to form the Modality Adaptation Pre-training Data with simple instructions of automatic speech recognition. And, for cross-modal instruction tuning, we use several open-source language-only instruction-tuning data sets to build the Cross-modal Instruction Fine-Tuning Data by generating the speech data. The details are as follows.

Modality Adaptation Pre-training Data. To align the embeddings of text and audio, we collect several public ASR data sets in both English and Chinese, including Aishell [24], LibriSpeech [25], Magicdata [26] and Primewords [27]. The data sample of ASR data usually consists of a pair of speech audio and text utterances, especially, when we add a simple instruction to the data sample as the task instruction. These simple instructions are listed in Figure 2 and Figure 3, which are different representations of the automatic speech recognition task in both English and Chinese. While pre-training, the simple instruction and the audio data are input to the model to predict the text label of the audio data.

X_{sample} = B_INST + B_SYS + SYSTEM + E_SYS + $\{content_{user}\}$ + E_INST + $\{content_{gpt}\}$

$content_{user}$ = $\{audio_{token}\}$ + "\n" + I_{simple} (or $content_{user}$ = I_{simple} + "\n" + $\{audio_{token}\}$)

$content_{gpt}$ = TEXT_LABEL

$audio_{token}$ = AUDIO_START_TOKEN + AUDIO_PATCH_TOKEN * {audio_token_len} + AUDIO_END_TOKEN

Figure 4: The sample sequence format for the pre-training. We follow the manner of Llama-2, and B_INST = '[INST]', E_INST = '[/INST]', B_SYS = '<<SYS>> \n', E_SYS = '$\backslash n$ <</SYS>> \n\n'. The SYSTEM = 'You are a helpful language and speech assistant. You are able to understand the speech content that the user provides, and assist the user with a variety of tasks using natural language.', and the TEXT_LABEL is the text label of the ASR data sample. The audio_token_len is set to 64 by default. Special audio tokens are used, AUDIO_START_TOKEN = '<au_start>', AUDIO_END_TOKEN = '<au_end>', AUDIO_PATCH_TOKEN = '<au_patch>'. The $content_{user}$ consists of the $audio_{token}$ and the I_{simple}, in which I_{simple} is a simple instruction and is randomly put before or after the $audio_{token}$. While training the BOS token and the EOS token will be added to each sample at the beginning and the end of the sequence, only the green tokens are used to compute the loss.

$X_{\text{instruction}}$ = <BOS> + X_{sample} + <EOS> + X_{sample} + <EOS> + ··· + <EOS>

X_{sample} = B_INST + B_SYS + SYSTEM + E_SYS + $\{content_{user}\}$ + E_INST + $\{content_{gpt}\}$

$content_{user}$ = $\{audio_{token}\}$, $content_{gpt}$ = TEXT_RESPONSE

$audio_{token}$ = AUDIO_START_TOKEN + AUDIO_PATCH_TOKEN * {audio_token_len} + AUDIO_END_TOKEN

Figure 5: The sample sequence format for the cross-modal instruction fine-tuning. We follow the manner of Llama-2, and B_INST = '[INST]', E_INST = '[/INST]', B_SYS = '<<SYS>> \n', E_SYS = '$\backslash n$ <</SYS>> \n\n'. The SYSTEM = 'You are a helpful language and speech assistant. You are able to understand the speech content that the user provides, and assist the user with a variety of tasks using natural language.', and the TEXT_RESPONSE is the text response from the chatbot. The audio_token_len is set to 64 by default. Special audio tokens are used, AUDIO_START_TOKEN = '<au_start>', AUDIO_END_TOKEN = '<au_end>', AUDIO_PATCH_TOKEN = '<au_patch>'. The $content_{user}$ is the $audio_{token}$ which will be replaced by the audio embeddings during training. Each round of question and answer will be formatted as X_{sample}, which will be concatenated together with the EOS token. While training the BOS token will be added at the beginning of the sequence, and the EOS token will be added at the end of the sequence, only the green tokens are used to compute the loss.

Cross-modal Instruction Fine-Tuning Data. As the effectiveness of the open-source language-only instruction-tuning data sets has been demonstrated in previous works[15, 16, 17], a natural idea is to generate audio data of these language-only data sets to build a cross-modal instruction-tuning data. In the process of building this dataset, we first carefully filtered all the conversation data, by removing the conversations that are not suitable for vocalization, including codes, a large number of symbols, URLs, and other non-readable text. To ensure the data quality, in the second stage, all the answers from chat-bots in the conversations are filtered again. Those that do not contain valuable information are dropped. In the third stage, we use Microsoft Azure text-to-speech API [28] to generate speech data from humans in these data sets. The speech data of humans are used as the complex instructions and the responses from the chatbot are predicted during the instruction fine-tuning. Specifically, 80k conversation data which contains 160k samples is selected from WizardLM [17], 23k conversation data which contains 155k samples is selected from ShareGPT [16] and 96k conversation data which contains 192k samples is selected from GPT-4-LLM [15]. Table 1 shows the specific details of the dataset, which contains 199k conversation data and 508k samples in total. Several examples of the dataset are shown in Figure 6. We release this dataset as LLaSM-Audio-Instructions at `https://huggingface.co/datasets/LinkSoul/LLaSM-Audio-Instructions`.

Table 1: LLaSM-Audio-Instructions Data.

LLaSM-Audio-Instructions				
Source	Conversations	Samples	English Samples	Chinese Samples
WizardLM	80k	160k	159k	<1k
ShareGPT	23k	155k	140k	15k
GPT-4-LLM	96k	192k	128k	64k
Total	199k	508k	428k	80k

What are the three primary colors?

Human: /dataset/1038/1.mp3

Agent: The three primary colors are red, blue, and yellow. These colors are called primary because they cannot be created by mixing other colors and all other colors can be made by combining them in various proportions. In the additive color system, used for light, the primary colors are red, green, and blue (RGB).

帮我找一个适合我兄弟的礼物。他是一个狂热的体育迷。

Human: /dataset/19791/1.mp3

Agent: 根据您的描述，您的兄弟是一名狂热的体育迷。对于这样的人来说，有一些很棒的礼物选择，包括：

1. 球队服装：例如，如果他是一名足球迷，您可以为他购买他最喜欢球队的球衣或帽子。
2. 运动装备：根据他喜欢的运动，您可以为他购买一些相关的运动装备，如篮球、足球、棒球等。
3. 球票：如果可能的话，您可以为他购买他最喜欢的体育比赛的门票，让他有机会亲身感受现场的激情与刺激。
4. 运动类书籍/杂志/纪录片：您可以为他购买一些关于运动的书籍、杂志或纪录片，让他能够更深入地了解他喜欢的运动。
5. 运动类游戏：您可以为他购买一些运动类电子游戏，让他能够在家中也能享受运动的乐趣。

希望这些建议能够帮到您，为您的兄弟选购一份他喜欢的礼物。

How are cellular networks tested in labs for performance?

Human: /dataset/1203/1.mp3

Agent: Cellular networks are typically tested in labs using specialized equipment such as cellular network simulators and test mobile devices. The network simulator generates simulated network traffic and conditions, while the test mobile devices are used to measure the performance of the network under test. The performance metrics that are typically measured include throughput, latency, signal strength, and error rates. Additionally, various tests such as drive tests and walk tests are also performed to measure network perfo scena

What are drive tests in context of cellular networks?

Human: /dataset/1203/3.mp3

Agent: Drive tests, also known as “drive-by testing”, are a type of field test that is used to measure the performance of a cellular network while a vehicle is in motion. The test vehicle is equipped with specialized test equipment, such as a data logger and a test mobile device, which is used to measure the performance of the network as the vehicle travels through different areas of the network's coverage. The test vehicle typically drives along a pre-determined route, and the test equipment is used to collect data on key performance indicators such as signal strength, call quality, and data throughput. This data is then analyzed to identify areas of the network that may be experiencing poor performance or capacity issues. Drive tests are useful for measuring the performance of cellular networks in real-world conditions and can help network operators identify and resolve issues related to network coverage, capacity, and quality.

Figure 6: Data samples of the LLaSM-Audio-Instructions.

Figure 7: Examples of experiments.

4 Experiments

As shown in Figure 7, our proposed model, LLaSM, can adaptively recognize and respond to speech in Chinese and English. Figure 7 further demonstrates the effectiveness of LLaSM in a bilingual setting. Unlike conventional models that rely on speech-to-text conversion as a preprocessing step, LLaSM can directly process speech inputs, which improves its execution efficiency. Furthermore, LLaSM can support multiple languages and scenarios, which expands its application range. Therefore, LLaSM is a promising model for convenient and interactive human-artificial intelligence communication.

5 Conclusion

This work presents LLaSM, a large language model with cross-modal conversational abilities, capable of understanding and following speech-and-language instructions. Experiments show that LLaSM demonstrates a more convenient and natural way for humans to interact with artificial intelligence. Specifically, to alleviate the scarcity of cross-modal speech-and-language instructions data, we build a large Speech Instruction Following data set LLaSM-Audio-Instructions. It is the largest Chinese and English speech-text cross-modal instruction-following data set to our knowledge. Finally, by adopting a visual modal encoder that can easily provide LLaSM with visual capabilities, we will explore combining both vision and audio modalities in future work.

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